

(or used to) for our photos. After this, the canned and processed film would be edited to create the movie, which would be output and duplicated for projection at movie theatres.

While some parts of this process still remain, there are new stages that have been added to this workflow, if you will, which are dramatically altering the way a movie looks.

Take, for example, the recent blockbuster *Black*. The entire soft, dreamy look the movie has could not have been possible using just the camera. Enter the post-production experts. Using software such as AutoDesk Lustre, the entire movie (after editing) was given a look and feel that was consistent throughout, and did not cause the viewer to think there was something 'special' going on. This, however, is a special case. What is not a special case, though, is the Digital Intermediate (DI) that virtually all movies today undergo.

Digital Intermediate

DI is an ever-evolving process. In a nutshell, it is the middle one-third of the operation loosely termed 'Digital Film'. The other two-thirds of Digital Film consist of acquisition (the obtaining of the source material through some form of capture) and presentation (the distribution, projection and/or transmission of the final result).

But before DI comes acquisition and editing. The entire movie is usually transferred to a computer and digitised, making it easier to edit. In some cases, though, filmmakers prefer to edit in the linear format—from tape to tape, without transferring it to a computer or digitising it.

In either case, the DI process begins after editing is complete. The transferred film is colour-corrected, 'cleaned up' (more on this later) and composited (put together again as one package). The final product is then output to either analogue (film) format or digital format for projection.

What happens in DI is, colours are made consistent, edit jerks are cleaned, and special effects—even the basic ones—are added. Remember the multiple screens in *Dhoom* where four different shots were visible at the same time? All four were shot separately and were made to look consistent using DI.

But DI is going beyond post-production: it is now being used in pre-production. "*Dhoom 2* will be using a pre-viz (pre-visualisation) technique that will create a blueprint of all shots. This will mean that everyone on the set knows exactly what is to be done before they even reach the location. The cost saving and clarity of job responsibility this offers is phenomenal," says Pankaj Kedia, Regional Manager, South East Asia and India, Autodesk (Media and Entertainment Division).

Indeed, one of the greatest advantages DI and other such processes have offered is the cost savings. At the same time, these have also resulted in increased creativity and better planning—which has, in turn, has meant that less time is spent wondering what to do when on the sets.

"Tools such as pre-viz are finally being used. These allow the director to decide where everyone in the unit will be for a shot. Everyone involved, too, is aware of this, and that makes for less time spent on the sets," says Pankaj Khandpur, Creative Director Visual Computing Labs, Tata Elxsi.

So Will Tech Take Over?

Some time ago, myths circulated as the use of technology increased: for example, that cinematographers might go obsolete, or that they would not have to work as hard. But these were never true. "Yes, there were a lot of ideas that now cinematographers will not have to work as hard, when these processes were gaining popularity in the US around 1990. But this is definitely not true," assures Sibille Cooney, Senior Trainer, Autodesk M&E Division, US. Cooney is, in fact, engaged in training upcoming Indian editors in the art of DI and post-production effects.

Agreeing with Cooney is one of India's leading post-production and Visual Effects (VFX) artists, Huzefa Lokhandwala of Prime Focus Studios. "Each aspect of the post-production process sometimes requires varied specialised training. Hands-on experience thereafter definitely helps. Not speaking technically, however, we could safely say that the basic requirement would be the ability to observe and visualise. The ability to combine man and machine in the most effective and aesthetic fashion is what makes a good post-production artist in this technological age," he says.

